

AFRILANG Stdlib

std/c/sigconv

Bibliothèque AFRILANG `std/c/sigconv` (niveau complex, catégorie complex). Elle expose 6 fonction(s) : convolve, crossCo

```
`import "std/c/sigconv" . `use sigconv`
```

- `convolve(a list number, b list number) ? list number`
- `crossCorr(a list number, b list number) ? list number`
- `movingAvg(v list number, win number) ? list number`
- `deconvSimple(signal list number, kernel list number) ? list number`
- `energy(v list number) ? number`
- `normalize(v list number) ? list number`

Category: complex | Tier: complex | Functions: 6

FRANCAIS

1. Vue d'ensemble

Bibliothèque AFRILANG ``std/c/sigconv`` (niveau complex, catégorie complex). Elle expose 6 fonction(s) : `convolve`, `crossCo`

```
`import "std/c/sigconv"` · `use sigconv`  
- `convolve(a list number, b list number) ? list number`  
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- `energy(v list number) ? number`  
- `normalize(v list number) ? list number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/sigconv"  
use sigconv
```

Niveau : complex · Catégorie : Modules complex (c/)

Domaines avancés ? graphes, NLP, réseaux complexes.

3. Référence API

```
? convolve(a list number, b list number) ? list  
? crossCorr(a list number, b list number) ? list  
? movingAvg(v list number, win number) ? list  
? deconvSimple(signal list number, kernel list number) ? list  
? energy(v list number) ? number  
? normalize(v list number) ? list
```

4. Exemples d'utilisation

```
import "std/c/sigconv"  
use sigconv  
  
create result = convolve(/* args */)   
say result  
  
create result = crossCorr(/* args */)   
say result  
  
create result = movingAvg(/* args */)   
say result  
  
create result = deconvSimple(/* args */)   
say result  
  
create result = energy(/* args */)   
say result  
  
create result = normalize(/* args */)   
say result
```

5. Bonnes pratiques

```
? Préférez `import "std/c/sigconv"` en tête de fichier.  
? Lancez `afrilang check` avant de livrer.  
? Combinez avec les modules stdlib de la même catégorie.  
? Voir /stdlib/ pour le source live et les démos playground.  
? Signalez les problèmes sur GitHub si une export manque.
```

6. Annexe ? source

```
module sigconv  
  
function convolve(a list number, b list number) returns list number  
end
```

```
function crossCorr(a list number, b list number) returns list number
end

function movingAvg(v list number, win number) returns list number
end

function deconvSimple(signal list number, kernel list number) returns list number
end

function energy(v list number) returns number
end

function normalize(v list number) returns list number
end
end
```

ENGLISH

1. Overview

AFRILANG library `std/c/sigconv` (tier complex, category complex). Exports 6 function(s):
convolve, crossCorr, movingAvg

```
`import "std/c/sigconv" . `use sigconv`  
- `convolve(a list number, b list number) ? list number`  
- `crossCorr(a list number, b list number) ? list number`  
- `movingAvg(v list number, win number) ? list number`  
- `deconvSimple(signal list number, kernel list number) ? list number`  
- `energy(v list number) ? number`  
- `normalize(v list number) ? list number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/sigconv"  
use sigconv
```

Tier: complex · Category: Complex modules (c/)
Advanced domains ? graphs, NLP, complex networks.

3. API reference

```
? convolve(a list number, b list number) ? list  
? crossCorr(a list number, b list number) ? list  
? movingAvg(v list number, win number) ? list  
? deconvSimple(signal list number, kernel list number) ? list  
? energy(v list number) ? number  
? normalize(v list number) ? list
```

4. Usage examples

```
import "std/c/sigconv"  
use sigconv  
  
create result = convolve(/* args */)   
say result  
  
create result = crossCorr(/* args */)   
say result  
  
create result = movingAvg(/* args */)   
say result  
  
create result = deconvSimple(/* args */)   
say result  
  
create result = energy(/* args */)   
say result  
  
create result = normalize(/* args */)   
say result
```

5. Best practices

```
? Prefer `import "std/c/sigconv"` at the top of your file.  
? Run `afirang check` before shipping.  
? Combine with related stdlib modules from the same category.  
? See /stdlib/ for the live source and playground demos.  
? Report issues on GitHub if an export is missing or undocumented.
```

6. Appendix ? source

```
module sigconv  
  
function convolve(a list number, b list number) returns list number  
end
```

```
function crossCorr(a list number, b list number) returns list number
end

function movingAvg(v list number, win number) returns list number
end

function deconvSimple(signal list number, kernel list number) returns list number
end

function energy(v list number) returns number
end

function normalize(v list number) returns list number
end
end
```